Domain and Range

Section 1: Finding Domains of Rational and Radical Functions

1 Find the domain of $f(x) = \frac{x-3}{x^2-16}$.

a. $(-\infty, -4) \cup (-4, 4) \cup (4, \infty)$

b. $(-\infty, 4) \cup (4, \infty)$

c. $[-4, 4]$

d. All real numbers except $x = 3$

2 Find the domain of $g(x) = \sqrt{2x-8}$.

a. $[4, \infty)$

b. $(4, \infty)$

c. $[8, \infty)$

d. $(-\infty, 4]$

3 Find the domain of $h(x) = \frac{1}{\sqrt{9-x^2}}$.

a. $(-3, 3)$

b. $[-3, 3]$

c. $(-\infty, -3) \cup (3, \infty)$

d. $[0, 3)$

Section 2: Finding Ranges of Algebraic Functions

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4 Find the range of $f(x) = x^2 - 4x + 7$.

a. $[3, \infty)$

b. $[7, \infty)$

c. $(-\infty, 3]$

d. $[0, \infty)$

5 Find the range of $g(x) = -|x + 2| + 5$.

a. $(-\infty, 5]$

b. $[5, \infty)$

c. $(-\infty, -2]$

d. $[-5, 5]$

6 Find the range of $h(x) = \sqrt{25 - x^2}$.

a. $[0, 5]$

b. $[-5, 5]$

c. $[0, 25]$

d. $[5, \infty)$

Section 3: Piecewise Functions

7 Evaluate $f(-2)$ for $f(x)$ defined as $2x + 1$ for $x < 0$, and $x^2 - 3$ for $x \geq 0$.

a. -3

b. 1

c. -2

d. 7

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8 Find the domain of $f(x)$ defined as $\frac{1}{x+3}$ for $x < 1$, and $\sqrt{x-1}$ for $x \geq 1$.

a. $(-\infty, -3) \cup (-3, \infty)$

b. $[1, \infty)$

c. All real numbers

d. $(-\infty, 1) \cup [1, \infty)$

9 Determine the range of $g(x)$ defined as $-x$ for $x < 0$, and $x + 1$ for $x \geq 0$.

a. $(0, \infty)$

b. $[1, \infty)$

c. $[0, \infty)$

d. All real numbers

Section 4: Domain and Range of Compositions

10 If $f(x) = \frac{1}{x-2}$ and $g(x) = \sqrt{x}$, find the domain of $(f \circ g)(x)$.

a. $[0, 4) \cup (4, \infty)$

b. $[0, \infty)$

c. $(4, \infty)$

d. All reals except $x = 2$

11 If $f(x) = x^2$ and $g(x) = 3x - 1$, what is the range of $(f \circ g)(x)$ for $x \in \mathbb{R}$?

a. $[0, \infty)$

b. $[-1, \infty)$

c. $[1, \infty)$

d. All real numbers

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12 Find the domain of $(g \circ f)(x)$ if $f(x) = 2x + 6$ and $g(x) = \ln(x)$.

a. $(-3, \infty)$

b. $[0, \infty)$

c. $(-6, \infty)$

d. $(3, \infty)$

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